

## **DRAWING DESCRIPTIONS — PATENT C**

**Method and System for Autonomous Self-Regulation and Cognitive**

**Resynchronization in Neural Processing Systems During Disconnection**

**from External Control Layers**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

**FIG. 1 is a system architecture block diagram showing the neural processing system with the cognitive control layer, fallback controller, and the transition mechanism between connected and autonomous operation.**

**FIG. 2 is a state transition diagram showing the three operational states and the conditions that trigger transitions between them.**

**FIG. 3 is a graph showing the energy-aware modulation curve that governs autonomous self-regulation behavior across four energy regions.**

**FIG. 4 is a signal diagram showing the output waveform of the autonomous**

input generator including circadian base signal and attention bursts.

**FIG. 5 is a structural diagram of the resynchronization payload transmitted to the cognitive control layer upon reconnection.**

**FIG. 6 is a set of timeline diagrams showing three recovery scenarios: normal reconnection, crash recovery, and clean shutdown.**

**FIG. 7 is an end-to-end signal flow diagram showing the complete system during both connected and autonomous operation, illustrating how the fallback mechanism seamlessly replaces external cognitive modulation with internal energy-aware self-modulation without interrupting neural processing or energy harvesting.**

## **DETAILED DESCRIPTION OF THE DRAWINGS**

### **FIG. 1 — System Architecture with Fallback**

**This figure shows the complete system architecture including the fallback mechanism.**

**Top section — Cognitive Control Layer:**

- Cloud shape labeled "External Cognitive Control Layer (e.g., Claude)"
- Two arrows exiting downward:
- "Modulation commands (-1.0 to +1.0)"
- "Heartbeat signal (implicit via tool calls)"
- Label: "Provides intelligent cognitive decisions"
- Dashed outline indicating "May become unavailable"

**Middle section — Transition Controller:**

- Rectangle labeled "Heartbeat Watchdog"
- Inside: "Monitors time since last cognitive input"
- Inside: "Timeout threshold: 30 seconds"
- Inside: "Check interval: 5 seconds"
- Two output paths:

- Left arrow (solid): "Heartbeat active -> Connected Mode"

- Right arrow (dashed): "Heartbeat timeout -> Autonomous Mode"

**Left path — Connected Mode:**

- Rectangle labeled "Cognitive-Driven Operation"

- Inside: "Input: Claude-controlled (0-1)"

- Inside: "Modulation: Claude-controlled (-1 to +1)"

- Arrow down to SNN

**Right path — Autonomous Mode:**

- Rectangle labeled "Autonomous Fallback Controller"

- Inside: "Input: Self-generated (circadian + bursts)"

- Inside: "Modulation: Energy-aware (-0.4 to +0.3)"

- Arrow down to SNN

- Additional outputs:

- Arrow to "Step Buffer" (rectangle)

- Arrow to "State Snapshots" (cylinder/disk shape)

**Bottom section — Neural Processing:**

- Large rectangle labeled "Spiking Neural Network + Energy Harvester"

- Inside: "Continuous operation regardless of control source"

- Label: "THE SYSTEM NEVER STOPS"

**Key annotation:**

- "Seamless transition: SNN continues processing without interruption"

- "The control source changes; the computation continues"

**FIG. 2 — State Transition Diagram**

This figure shows a finite state machine with three states and labeled transitions.

**Three states (large circles or rounded rectangles):**

**State 1 (left, bold outline):**

- **Label: "CONNECTED"**
- **Sublabel: "Cognitive layer active"**
- **Inside: "Input from cognitive layer"**
- **Inside: "Modulation from cognitive layer"**

**State 2 (right):**

- **Label: "AUTONOMOUS"**
- **Sublabel: "Fallback active"**
- **Inside: "Self-generated input"**
- **Inside: "Energy-aware modulation"**
- **Inside: "State buffering active"**

**State 3 (bottom):**

- **Label: "RECOVERING"**

- Sublabel: "Resynchronization"

- Inside: "Transmitting buffer"

- Inside: "Restoring cognitive control"

Transitions (labeled arrows between states):

**CONNECTED -> AUTONOMOUS:**

- Arrow labeled: "Heartbeat timeout (>30s no tool call)"

- Condition: "last\_tool\_call\_time > 30s"

**AUTONOMOUS -> RECOVERING:**

- Arrow labeled: "Cognitive layer reconnects (resync called)"

- Condition: "resync() tool invoked"

**RECOVERING -> CONNECTED:**

- Arrow labeled: "Resync complete, buffer transmitted"

- Condition: "Cognitive layer acknowledges state"

**AUTONOMOUS -> AUTONOMOUS (self-loop):**

- **Arrow labeled: "Each autonomous step (up to 10,000)"**

- **Condition: "autonomous\_steps < hard\_cap"**

**AUTONOMOUS -> CONNECTED (direct, dashed):**

- **Arrow labeled: "Any tool call received (heartbeat reset)"**

- **Condition: "Direct reconnection without explicit resync"**

**Additional annotations:**

- **Start indicator pointing to CONNECTED: "Initial state"**

- **Note on AUTONOMOUS: "Snapshot every 50 steps"**

- **Note on AUTONOMOUS self-loop: "Hard cap: 10,000 steps"**

### **FIG. 3 — Energy-Aware Modulation Curve**

This figure shows the step function mapping energy level to modulation value.

**X-axis: "System Energy Level (mWh)" ranging from 0 to 100+**

**Y-axis: "Autonomous Modulation Value" ranging from -0.5 to +0.5**

**Step function with four plateaus:**

**Plateau 1 (0 to 15 mWh):**

- **Modulation = -0.4**
- **Region shaded with dense hatching**
- **Label: "CRITICAL — Maximum conservation"**
- **Behavioral note: "Inhibit neural activity to preserve energy"**

**Plateau 2 (15 to 30 mWh):**

- **Modulation = -0.1**
- **Region shaded with light hatching**
- **Label: "LOW — Cautious operation"**
- **Behavioral note: "Slightly reduce activity"**

**Plateau 3 (30 to 80 mWh):**

- **Modulation = 0.0**

- **Region unshaded (clear)**

- **Label: "NORMAL — Neutral operation"**

- **Behavioral note: "Standard autonomous processing"**

**Plateau 4 (80+ mWh):**

- **Modulation = +0.3**

- **Region shaded with light hatching**

- **Label: "SURPLUS — Opportunistic exploration"**

- **Behavioral note: "Increase activity to utilize excess energy"**

Vertical dashed lines at energy thresholds (15, 30, 80 mWh) with labels.

**Annotation at bottom:**

- **"The system autonomously adjusts its processing intensity based on**

**available energy — mimicking biological metabolic regulation"**

#### **FIG. 4 — Autonomous Input Generator Output**

This figure shows a time-series waveform of the self-generated input signal.

**X-axis: "Autonomous Step Number" ranging from 0 to 500 (representative)**

**Y-axis: "Input Value" ranging from 0.0 to 1.0**

**Main waveform components:**

**Component 1 — Circadian Base Signal:**

- Smooth sinusoidal curve
- Equation annotation: " $\text{base} = 0.5 + 0.3 \times \sin(2 \times \pi \times \text{step} / \text{period})$ "
- Period labeled: "period = 100 steps"
- Amplitude range: 0.2 to 0.8

**Component 2 — Attention Bursts:**

- Sharp upward spikes superimposed on the base signal
- Occurring randomly at approximately 10% of steps

- Burst magnitude: +0.2 above base signal

- Each burst labeled with a small arrow: "10% probability burst"

### **Component 3 — Composite Signal:**

- The sum of base + bursts, shown as the final bold line

- Label: "Composite autonomous input"

**Below the main graph, a smaller graph showing the corresponding energy**

**level over the same time period:**

- Shows energy fluctuating based on activity

- Annotations showing where modulation changes (at 15, 30, 80 mWh thresholds)

- Label: "Energy-aware modulation applied to composite input"

### **Annotation:**

- "Circadian rhythm provides structured temporal variation"

- "Random bursts prevent the system from settling into trivial attractors"

- **"Energy-aware modulation adjusts overall intensity"**

### **FIG. 5 — Resynchronization Payload Structure**

This figure shows the hierarchical data structure of the resync payload.

**Top-level structure (large rectangle):**

- **Label: "Resynchronization Payload"**

**Three main sections within the payload:**

#### **Section 1 — Summary Statistics:**

- **Rectangle with fields:**

- **"steps\_autonomous: [integer]" — Number of steps run autonomously**
- **"energy\_delta: [float] mWh" — Net energy change during fallback**
- **"min\_energy: [float] mWh" — Minimum energy observed**
- **"max\_energy: [float] mWh" — Maximum energy observed**
- **"mean\_energy: [float] mWh" — Average energy level**

- "total\_spikes: [integer]" — Sum of all spikes during fallback

- "mean\_spikes\_per\_step: [float]" — Average spike rate

- Label: "ALWAYS included (compact summary)"

- Size annotation: "~200 bytes"

## Section 2 — Energy Delta Detail:

- Rectangle with field:

- "energy\_start: [float] mWh"

- "energy\_end: [float] mWh"

- "energy\_delta: [float] mWh"

- Arrow from energy\_delta to annotation: "Positive = system gained energy during autonomy"

## Section 3 — Full Step Buffer (Optional):

- Larger rectangle with repeating row structure:

- "Step 0: {input, modulation, spikes, energy, temp, pattern}"

- "Step 1: {input, modulation, spikes, energy, temp, pattern}"
- "..."
- "Step N: {input, modulation, spikes, energy, temp, pattern}"
- Label: "OPTIONAL — Full operational history"
- Size annotation: "~100 bytes x N steps"
- Condition: "Included when cognitive layer requests detailed history"

Two granularity levels annotated:

- Arrow to Section 1: "Level 1: Summary only (fast resync)"
- Arrow to Section 3: "Level 2: Full buffer (complete reconstruction)"

Decision flow at bottom:

- Diamond: "Cognitive layer requests full buffer?"
- Yes arrow: "Transmit Level 2 (summary + full buffer)"
- No arrow: "Transmit Level 1 (summary only)"

## FIG. 6 — Recovery Timeline Diagrams

This figure shows three horizontal timeline diagrams stacked vertically, each illustrating a different recovery scenario.

### Timeline A — Normal Reconnection:

Time -->

|-- Connected --|-- Timeout (30s) --|-- Autonomous --|-- Resync --|-- Connected -->

[Claude [Watchdog [Self-generated [Buffer [Cognitive

active] counting] input + energy transmitted] control

modulation] restored]

Labels on timeline:

- "Claude goes silent" (at transition to timeout)
- "Fallback engages" (at transition to autonomous)
- "resync() called" (at transition to resync)
- "Seamless resumption" (at return to connected)

- Buffer icon: "N steps buffered"

### **Timeline B — Crash Recovery:**

Time -->

|-- Connected --|-- CRASH --|-- Server Restart --|-- Snapshot Load --|-- Connected -->

[Normal [MCP server [Process [Reads [State

operation] terminates] restarts] latest.json] restored]

### **Labels on timeline:**

- "Server process dies" (at crash)

- "Process relaunched" (at restart)

- "initialize\_consciousness loads snapshot" (at snapshot load)

- "State restored from last snapshot (max 50 steps lost)" (annotation)

- Snapshot icon: "latest.json (serialized every 50 steps)"

### **Timeline C — Clean Shutdown:**

**Time -->**

**|-- Connected --|-- Autonomous --|-- EOF Signal --|-- Shutdown -->**

[Normal [Fallback [stdin closed] [Final snapshot

**operation] operation] written to disk]**

**Labels on timeline:**

- **"Claude disconnects" (at transition to autonomous)**

- **"Process receives EOF" (at EOF signal)**

- **"finally block writes snapshot" (at shutdown)**

- **"State preserved for next session" (annotation)**

**Common annotations for all three timelines:**

- **Solid line segments: "System actively processing"**

- **Dashed line segments: "System in transition"**

- **Bold marker at each state change: "No data loss at any transition"**

## FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous Operation

This figure shows the complete system signal flow in a split-view format, with connected operation on top and autonomous operation on bottom, sharing the same core processing pipeline.

Layout — Two parallel horizontal flows sharing a common center:

TOP HALF — Connected Operation:

Label in top-left corner: "CONNECTED MODE"

[External Cognitive Layer] — cloud shape at top

|

v (dashed arrow: "Cognitive modulation commands")

[Sensors] -> [SNN] -> [Cognitive Modulation] -> [Motor Actuator] -> [Energy Harvester]

^ |

|\_\_\_\_\_ Energy feedback (bold dashed) \_\_\_\_\_|

^

|\_\_\_\_\_ Self-observation feedback (dotted) \_\_\_\_ from SNN output

**Label on Cognitive Modulation block: "External AI sets reflection\_coeff, modulation"**

**Label: "Heartbeat active (tool calls within 30s)"**

**DIVIDING LINE — Horizontal dashed line across full width**

**Label on line: "DISCONNECTION EVENT (heartbeat timeout > 30s)"**

**Arrow pointing down: "Seamless transition — no interruption to SNN or Harvester"**

**BOTTOM HALF — Autonomous Operation:**

**Label in bottom-left corner: "AUTONOMOUS MODE"**

[Autonomous Input Generator] — rectangle replacing Sensors

**Label inside: "Circadian base signal"**

**Label inside: "10% stochastic attention bursts"**

|

**v**

[SNN] -> [Energy-Aware Self-Modulation] -> [Motor Actuator] -> [Energy Harvester]

^ |

|\_\_\_\_\_ **Energy feedback (bold dashed)** \_\_\_\_\_|

^

|\_\_\_\_\_ **Self-observation feedback (dotted)** \_\_\_\_ **from SNN output**

**Label on Self-Modulation block: "Internal energy-aware regulation"**

**Four modulation regions shown as small annotation:**

**"<15 mWh: -0.4 | <30 mWh: -0.1 | 30-80 mWh: 0.0 | >80 mWh: +0.3"**

**Additional components in autonomous mode:**

[State Buffer] — cylinder/database shape

**Arrow from SNN: "Every step buffered"**

**Label: "Rolling buffer (max 10,000 entries)"**

[Snapshot Writer] — disk shape

**Arrow from State Buffer: "Every 50 steps"**

**Label: "latest.json (persistent storage)"**

**RIGHT SIDE — Reconnection:**

**Arrow from bottom half back to top half**

**Label: "resync() called"**

[Resync Payload] — rectangle

**Label inside: "steps\_autonomous, energy\_delta, summary"**

**Label inside: "Optional: full step-by-step buffer"**

**Arrow up to External Cognitive Layer: "Cognitive layer catches up"**

**Shared components (drawn once, spanning both halves):**

**The SNN, Motor Actuator, and Energy Harvester are the SAME blocks**

**in both halves — drawn in the center spanning the dividing line**

**Label: "Core pipeline never stops"**

**Key annotations:**

**"The fallback system changes the INPUT SOURCE and MODULATION SOURCE"**

**"The core neural processing, motor actuation, and energy harvesting"**

**"continue WITHOUT INTERRUPTION through the transition"**

**Legend box:**

**"Solid arrows: Forward signal path"**

**"Bold dashed: Energy feedback loop"**

**"Dotted: Self-observation feedback loop"**

**"Thin dashed: Cognitive/modulation control path"**

**"The transition between modes changes only the control paths,"**

**"never the core processing paths"**

**Sheet numbering: "7/7" at top center**

**GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)**

**- Paper size: 8.5 x 11 inches (US Letter)**

- **Top margin: 1 inch**
- **Left margin: 1 inch**
- **Right margin: 5/8 inch**
- **Bottom margin: 3/8 inch**
- **Black lines on white background ONLY**
- **No color, no grayscale shading (use hatching patterns instead)**
- **Line weight: sufficiently heavy for adequate reproduction**
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/7", "2/7", etc.
- **No frames or borders around drawing area**
- **All text in figures must be at least 1/8 inch (3.2 mm) high**
- **Reference numerals must be used consistently across all figures**